Graph Neural Networks

Emanuel Sallinger

Representations

KG Embeddings
Widely-applied, large family of ML models

Logical Knowledge in KGs
Highly expressive, diverse family of logical models.

Graph Neural Networks
Using the KG structure *as* a neural network.

Data Models
Overview of data models in different communities

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Graph Neural Networks

Using the KG structure
as a neural network.

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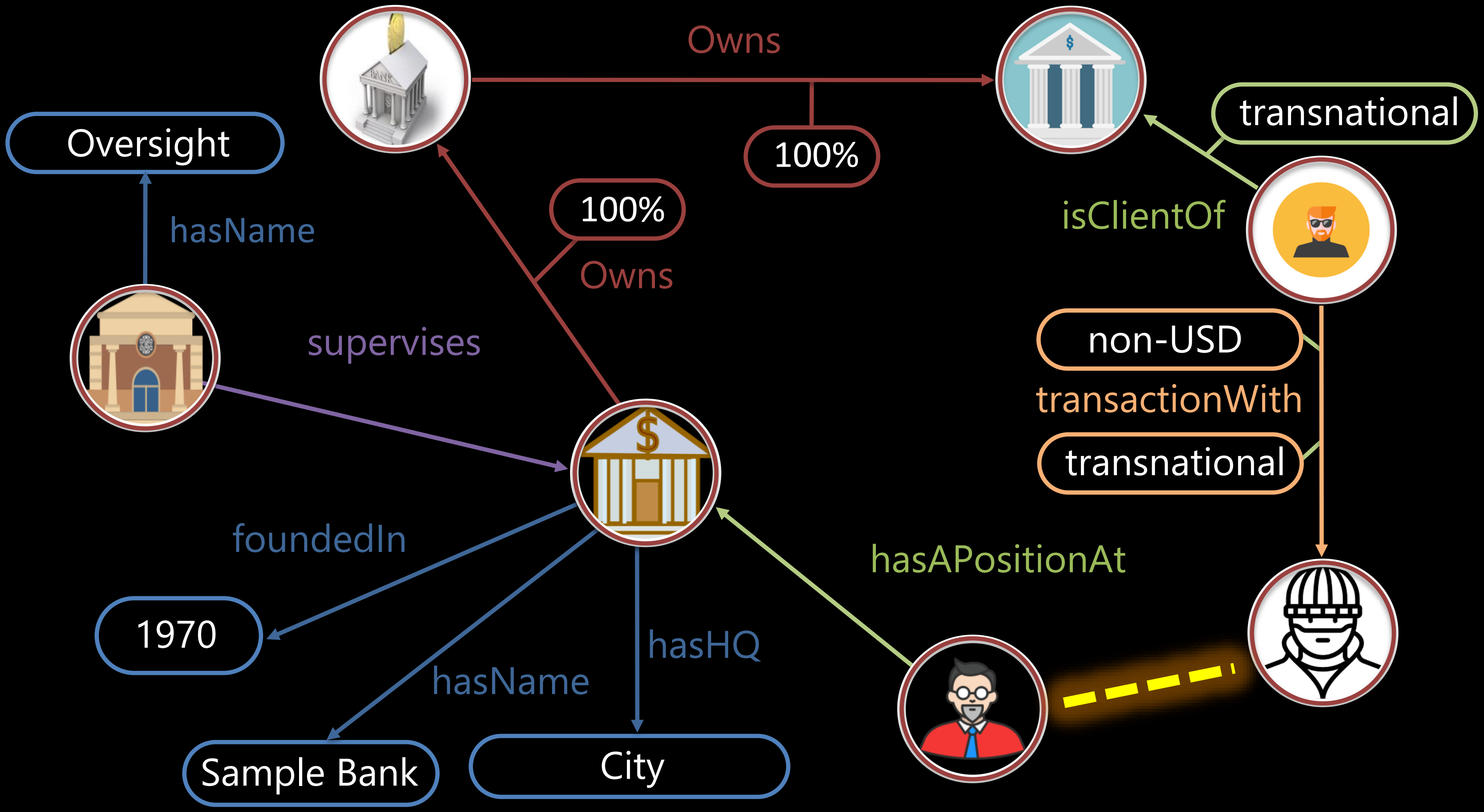
Owns
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Oversight
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isClientOf
Owns
supervises
non-USD
transactionWith
transnational
foundedIn
hasAPositionAt
1970
hasHQ
hasName
Sample Bank
City

Representations

Graph

Knowledge Graph
Data Models

Knowledge

Knowledge Graph
Embeddings

Logical
Knowledge

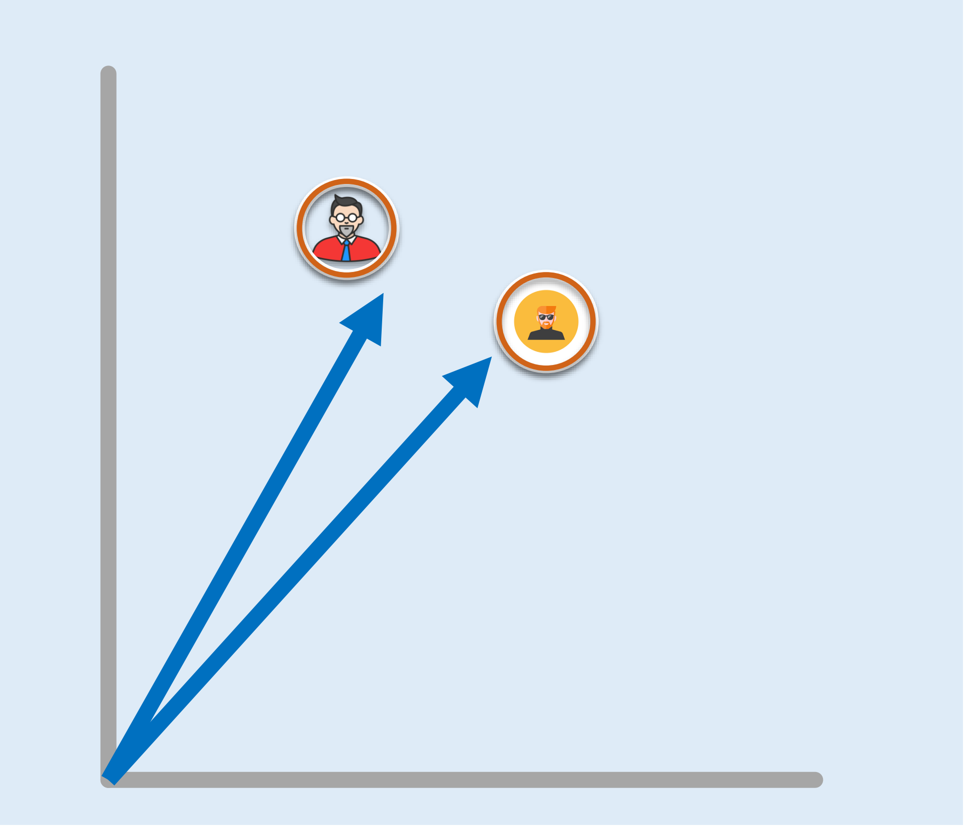
Graph Neural
Networks

Graph Neural Networks also Produce Embeddings!

$$e: \mathbb{S}_1 \rightarrow \mathbb{S}_2$$

$$e : \mathbb{V} \rightarrow \mathbb{R}^2$$

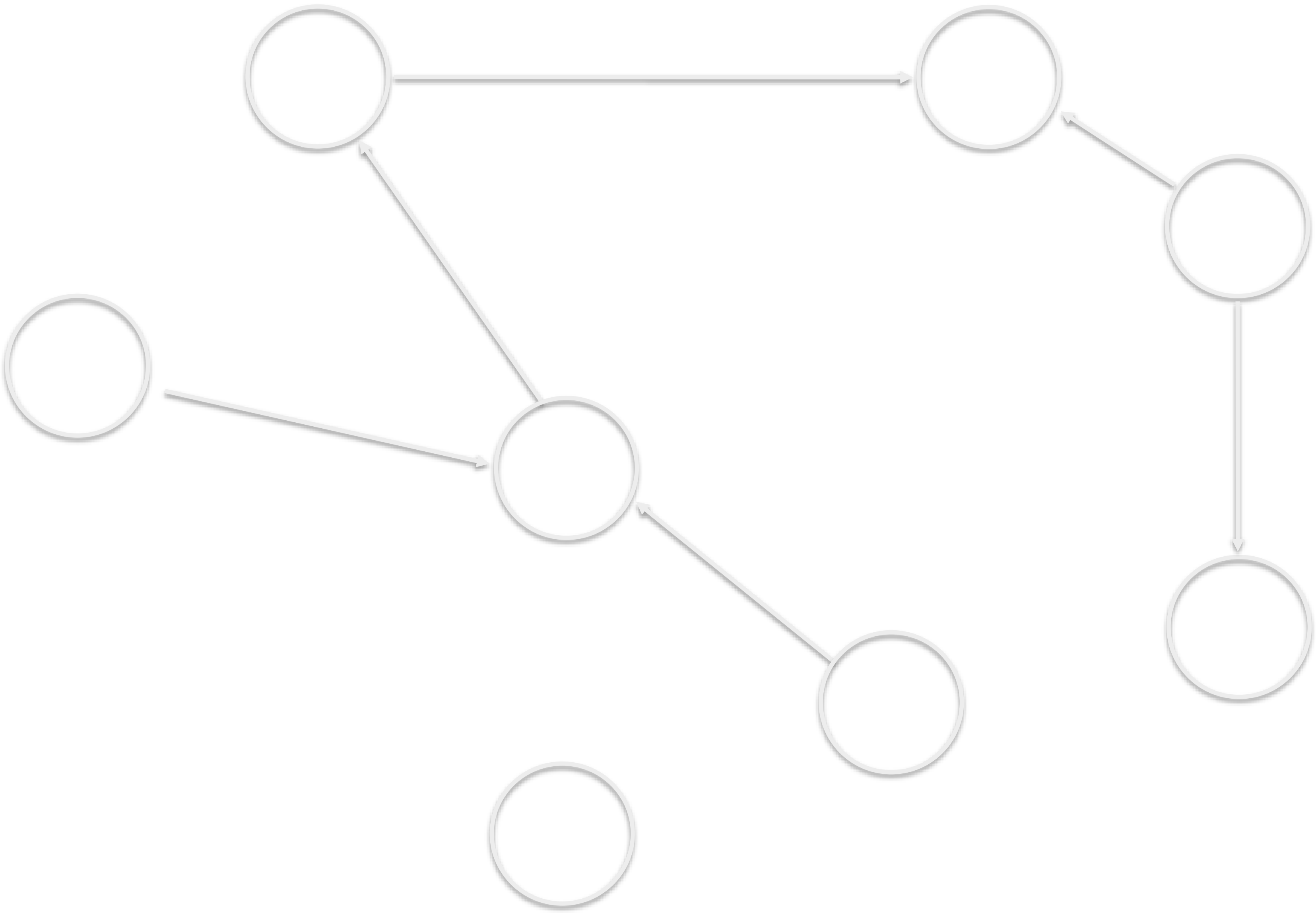
$\rightarrow$

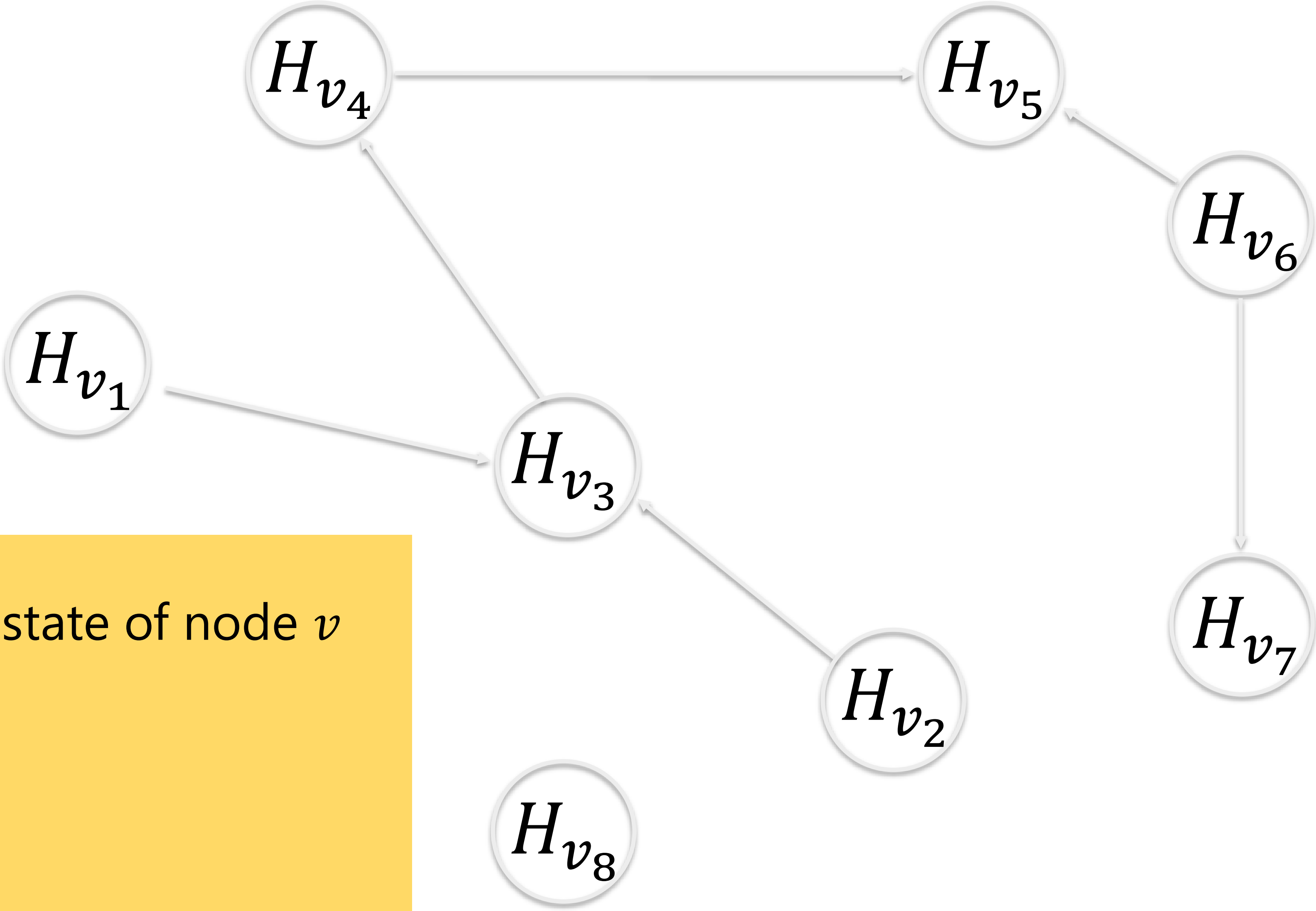


Main Differences

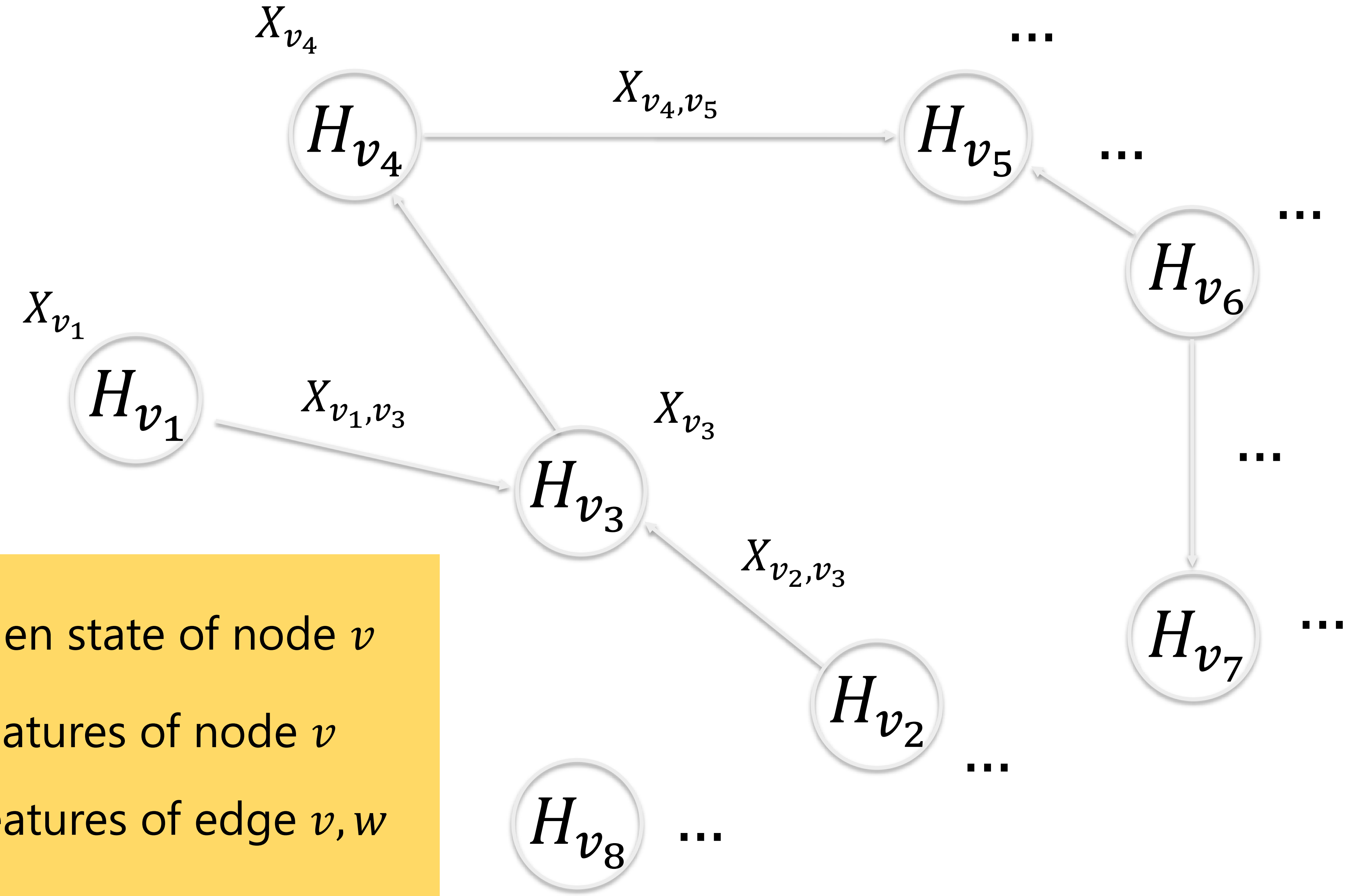
1. **Knowledge Graph Embeddings:**
   transductive: cannot be applied to unseen nodes

2. **Graph Neural Networks**
   inductive: can be applied to unseen nodes





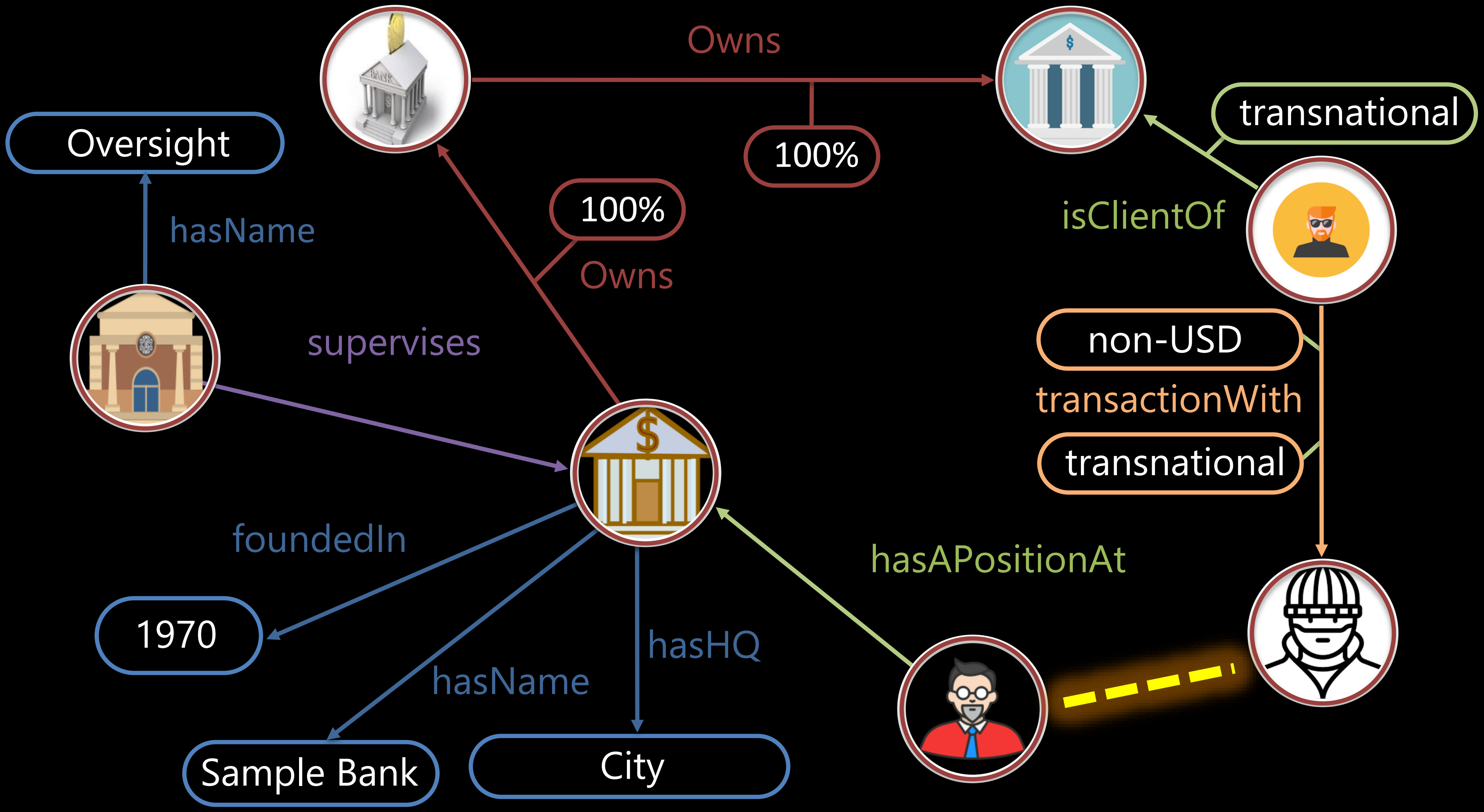
H_v hidden state of node v



H_v hidden state of node v

X_v features of node v

$X_{v,w}$ features of edge v,w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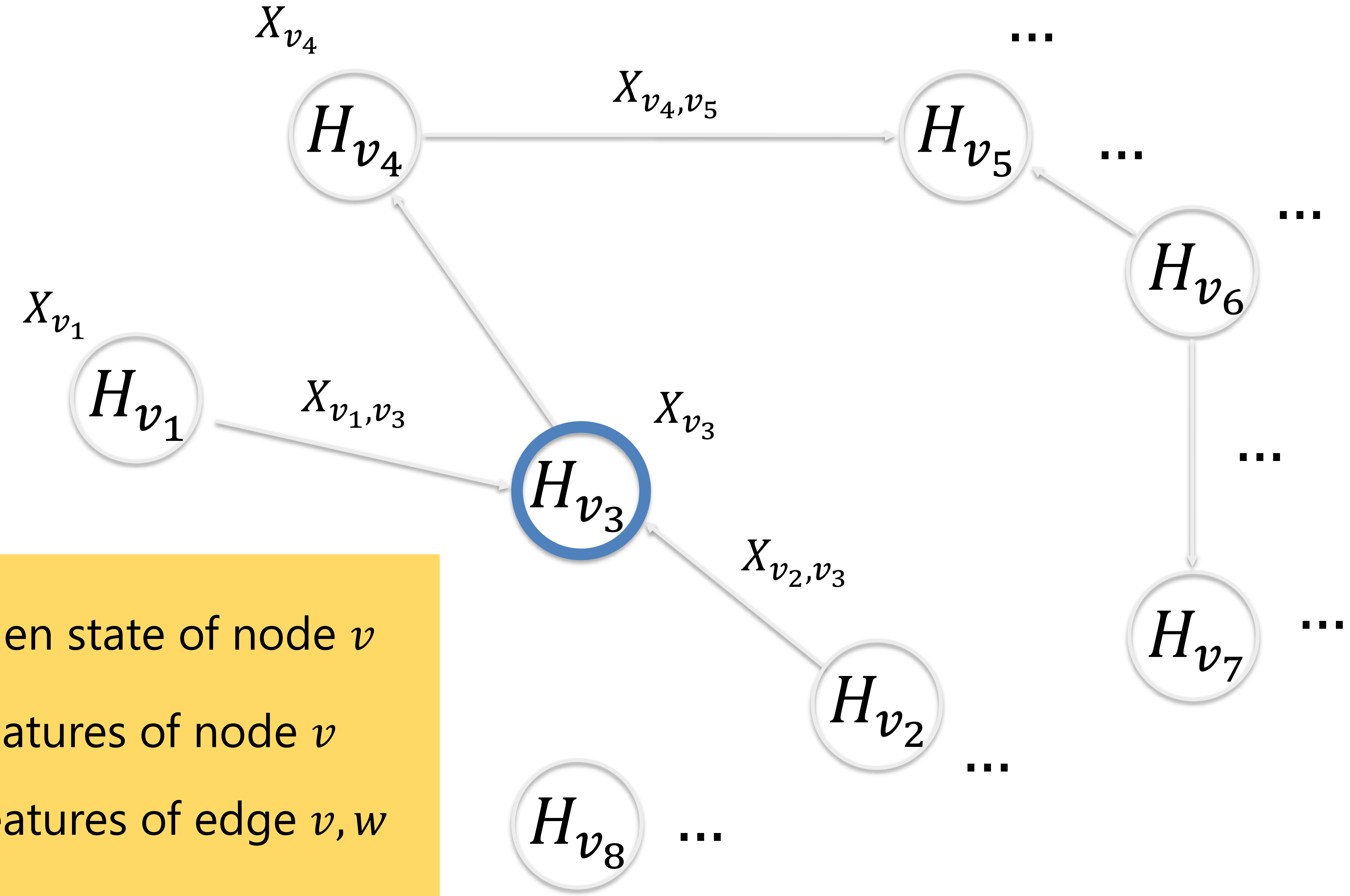
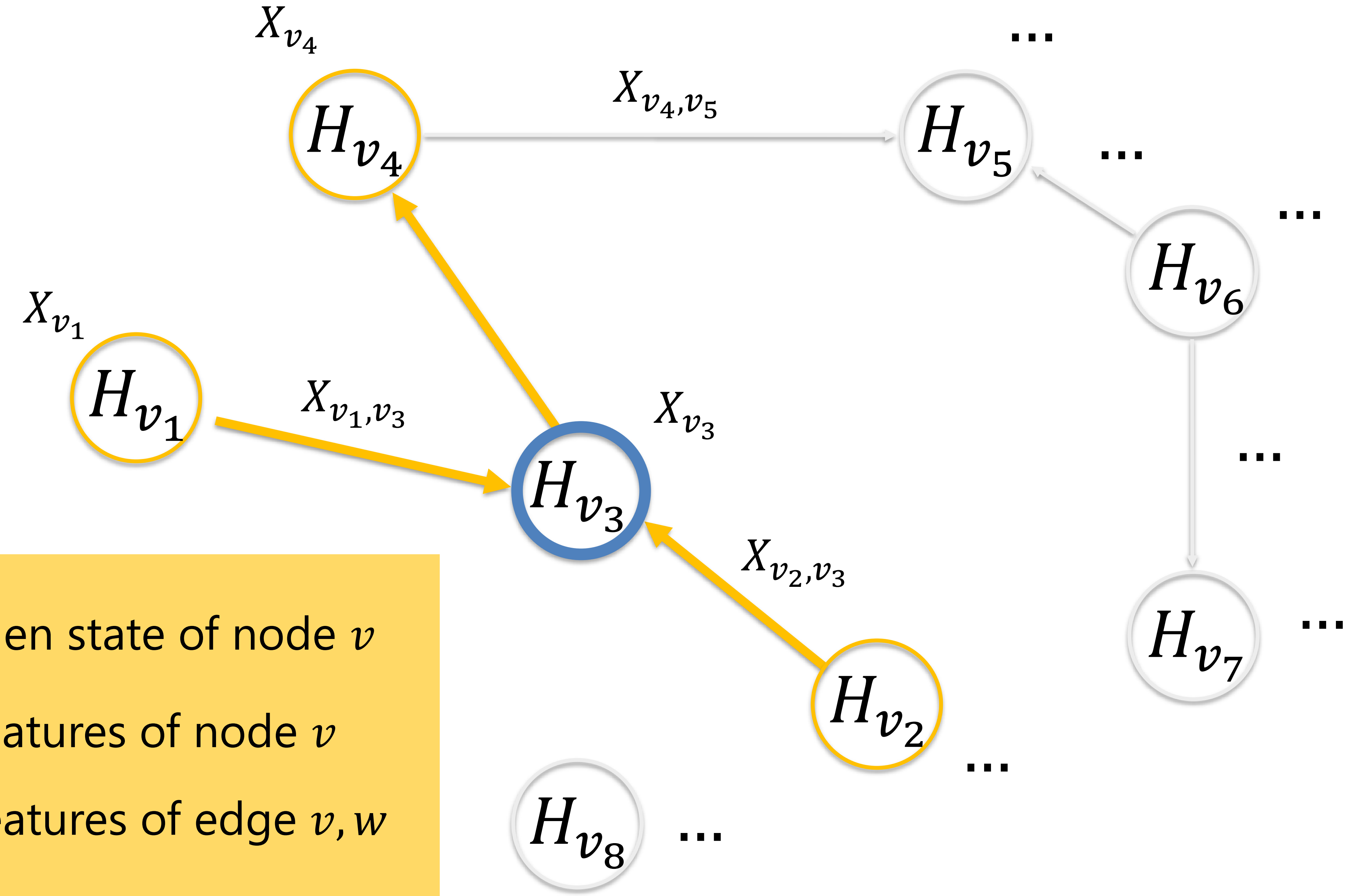


H_v hidden state of node v

X_v features of node v

$X_{v,w}$ features of edge v, w



H_v hidden state of node v

X_v features of node v

$X_{v,w}$ features of edge v, w

Graph Neural Networks

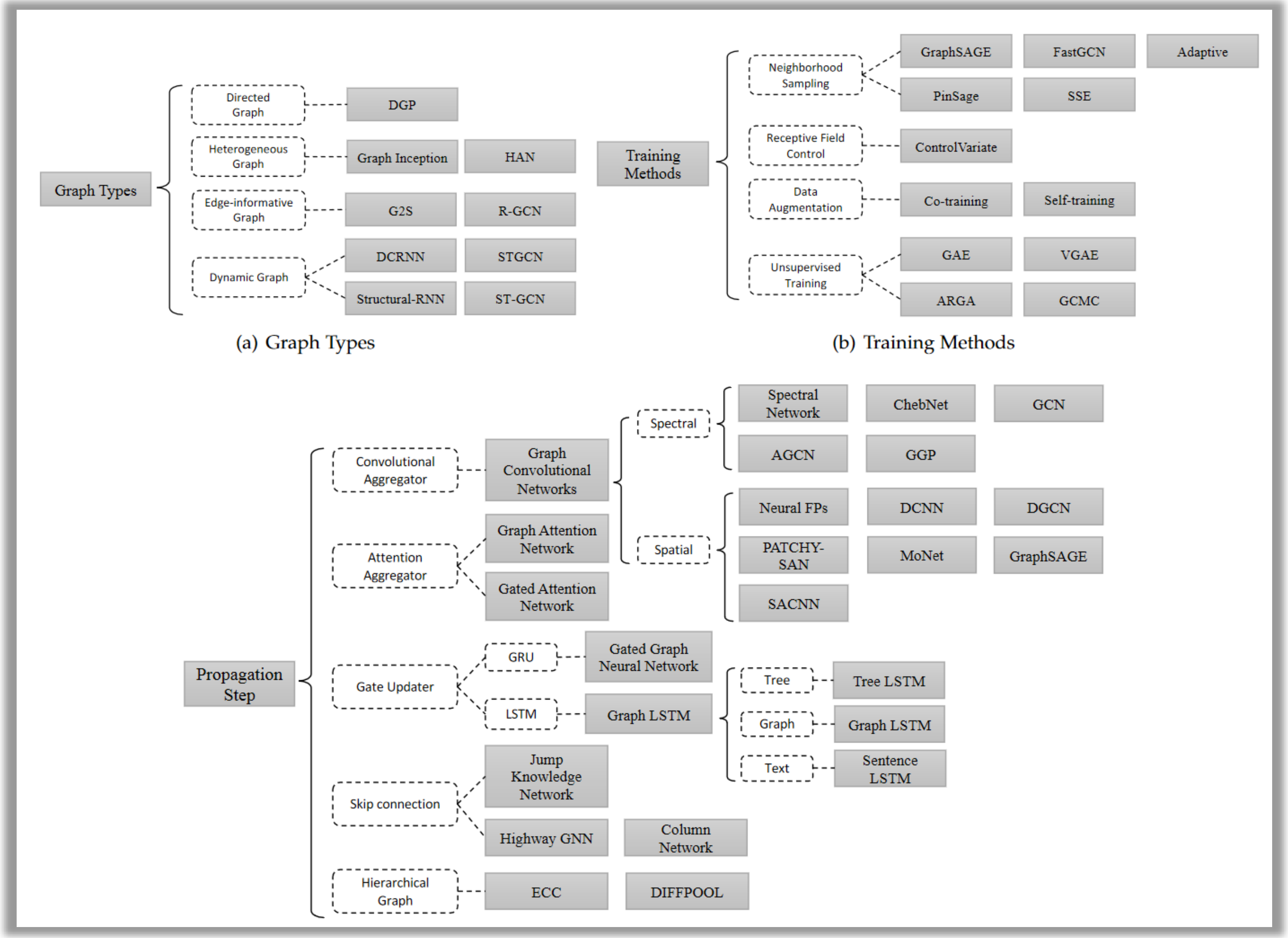
$$\boldsymbol{H} = F(\boldsymbol{H}, \boldsymbol{X})$$

F propagation step

Graph Neural Networks

$$\boldsymbol{H}^{t+1} = F(\boldsymbol{H}^{t}, \boldsymbol{X})$$

F propagation step



Source: https://arxiv.org/pdf/1812.08434.pdf

Graph Neural Networks

$$\boldsymbol{H} = F(\boldsymbol{H}, \boldsymbol{X})$$

F propagation step

Graph Neural Networks

$$\boldsymbol{H} = F_{\boldsymbol{W}}\left(\boldsymbol{H}, \boldsymbol{X}\right)$$

F propagation step

$\boldsymbol{W}$ Parameters (weights)

Graph Neural Networks:

Latent Knowledge

$$KG = (D, K; g, l, r)$$

$$\mathbb{K} = \mathbb{W} \text{ (w.r.t. } F\text{)}$$

$$l: \mathbb{D} \rightarrow \mathbb{W}$$

$$r_{\mathbb{W}}: \mathbb{D} \rightarrow \mathbb{D}$$

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